Scheme and answer key for

QP CODE: 25901448

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECPHY200 - ELECTRICAL CIRCUITS AND NETWORK SKILLS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 marks

Qn.	Question	Answer Key
1	Electrical energy consumed is given by	**a) V × I × Time**
2	In AC circuits, current is opposed by	**c) Resistance, Inductance and Capacitance**
3	Kirchhoff's Voltage Law applies to which of the following?	**b) A closed-circuit loop**
4	A galvanometer can measure only small currents. To use it as an ammeter, it must be	**c) Provided with a low resistance shunt in parallel**
5	What is the resultant resistance when two 2 Ω resistors are connected in series?	**c) 4 Ω**
6	The part of a transformer that carries magnetic flux is the:	**c) Core**
7	In a series circuit, the current is	**d) Same through all components**
8	The difference between synchronous speed and rotor speed is called	**d) Slip**

Qn.	Question	Answer Key
9	Surge is a	**b) Short voltage increase**
10	In delta connection, the wires are connected	**c) End to end**

Part B

Short Answer

Answer any 8 questions

Each question carries 5 mark

11. What is the physical meaning of real power in household appliances?

- Real power (P), measured in Watts (W), is the actual power dissipated by a load to perform useful work. It is the component where energy is transformed into another form.
- In appliances like irons or heaters, real power represents the heat generated. In motors, it represents the mechanical torque produced. It differs from reactive power, which merely cycles between the source and load.
- **Formula :** $P=V\ I\cos\phi$, where $\cos\phi$ is the power factor.

(any two relevant points 5 marks. 2.5 marks mark for 1 point)

12. Distinguish between Series and parallel connection.

Feature	Series Connection	Parallel Connection
Current Path	Single path	Multiple paths
Current (I)	Same in all components	Divided among branches
Voltage (V)	Divided among components	Same across all branches
Total Resistance	Increases $(R_s = \sum R)$ or	Decreases $(\frac{1}{R_p} = \sum \frac{1}{R})$

Feature	Series Connection	Parallel Connection
Failure Effect	Entire circuit stops	Only the failed branch stops

(any two relevant points 5 marks. 2.5 marks mark for 1 point)

13. Match with the best option.

- **(i) Energy → (D) Volt × Ampere × Time:** Energy is power multiplied by time. **(1 Mark)**
- **(ii) Temperature → (E) Resistance:** Resistance of most conductors changes with temperature. **(1 Mark)**
- **(iii) Current → (A) Charge:** Current is the rate of flow of electric charge. **(1 Mark)**
- **(iv) Voltage → (B) Potential difference:** Voltage is the work to move a unit charge. **(1 Mark)**
- **(v) Watt → (C) Power:** The SI unit of power. **(1 Mark)**

14. Distinguish between single-phase and three-phase AC sources.

- Single-phase uses one live and one neutral wire. Three-phase uses three live wires shifted by 120∘.
- Three-phase power is more efficient for transmitting electricity and provides non-pulsating power.
- Single-phase is used for domestic lighting (230V), while three-phase is for industrial motors (415V).

(any two relevant points 5 marks. 2.5 marks mark for 1 point)

15. Explain the response of a capacitor when connected to DC and AC sources.

- A capacitor blocks steady-state DC current after charging, acting as an open circuit.
- In AC, the capacitor continuously charges and discharges as polarity reverses, allowing current to "pass."
- The opposition is Capacitive Reactance:

$$X_C = \frac{1}{2\pi f C}$$

(any two relevant points 5 marks. 2.5 marks mark for 1 point)

16. Describe how the speed of a DC motor can be controlled.

Speed $N \propto \frac{V - I_a R_a}{\phi}$.

- Varying field current changes the magnetic flux (ϕ). Speed (N) is inversely proportional to flux ($N \propto 1/\phi$).
- Adding resistance in series with the armature reduces back EMF and speed.
- Varying supply voltage to the armature directly changes speed.

(any two relevant points 5 marks. 2.5 marks mark for 1 point)

17. What is inductance? Describe self-inductance and give the formula for inductive reactance.

- **Inductance (2 Mark):** The property of a conductor to oppose changes in electric current.
- **Self-Inductance (1.5 Marks):** Changing current in a coil induces an EMF in the *same* coil that opposes the change (Lenz's Law).
- **Inductive Reactance equation (1.5 Marks):**

$$X_L = 2\pi f L.$$

18. Explain the working principles of following Disconnect switches: a) knife switch b) Rotary switch c) isolator

- **a) Knife Switch (2 Marks):** A manual switch with a hinged metal blade. Closing the blade into the slot completes the circuit.
- **b) Rotary Switch (1.5 Marks):** Operates by rotating a central spindle to select between multiple circuit paths.
- **c) Isolator (1.5 Marks):** A mechanical device used to isolate a circuit for maintenance, operated under "no-load" conditions.

19. Describe the colour code of wires used in an extension board and explain significance.

- **Phase/Live (2 Marks):** Red or Brown. Carries current from source to appliance.
- **Neutral (1.5 Marks):** Black or Blue. Provides the return path to complete the circuit.
- **Earth/Ground (1.5 Marks):** Green or Yellow. Safety wire preventing shock by grounding fault currents.

20. Write a note on the role of insulators in cables.

- Prevents current leakage between conductors or to the environment.
- Provides a physical barrier preventing electric shocks to users.
- Protects internal strands from moisture, wear, and corrosion.

(any two relevant points 5 marks. 2.5 marks mark for 1 point)